

Applications of Microbiology with Biostatistics

Science

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Applications of Microbiology with Biostatistics (A27781)

Short Title: Applic Microbiol with Biostats
Faculty Science and Computing
Department Science
Cluster Biology
Author LCOFFEY
Credits 5

Level: Advanced

Description of Module / Aims

This module exposes the student to microbiological and molecular methods and their applications in a broad range of areas such as food, pharmaceutical and environmental. It also introduces the student to the use of microbes as analytical tools, and a range of topical and emerging microbiological issues and applications.

Programmes

BIOL-0027	BSc (Hons) in Molecular Biology with Biopharmaceutical Science (WD_SMBIO_B)	4 / 7 / M
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Indicative Content

- Review of culture-based microbiological methods for analysis of pharmaceuticals, foods, and environmental samples: use in applications such as sterility testing, bioburden, environmental monitoring, hygiene testing etc
- Nucleic acid-based methods for detection, identification and enumeration of micro-organisms, such as 16S rRNA gene sequencing, real-time PCR, PFGE, whole genome sequencing etc.
- Use of micro-organisms as analytical tools in applications such as clinical sample analysis, toxicity testing, MIC determination etc.
- Topical issues in microbiology, e.g. emerging diseases, drug discovery, the Human Microbiome Project etc.
- Topical epidemics and the human response e.g. Ebola, Zika
- Statistical interpretation of microbiological data

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Compare and contrast traditional and more modern microbiological methods.
2. Evaluate a wide range of applications of microorganisms and their products.
3. Evaluate appropriate analytical methods for microbiological analysis of specific samples.
4. Comment on new microbiological techniques and their future potential in industry.
5. Interpret statistical analysis of microbiology data.
6. Compare emerging pathogens and other topical microbiological issues.

Learning and Teaching Methods

- Lectures.
- Laboratory practicals
- Group discussions.
- Self-directed study.
- Formative assessment.
- Peer-learning.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	60%	1,4,5,6
Continuous Assessment	40%	
<i>Lab Report</i>	40%	2,3

Assessment Criteria

<40%: No understanding of applications of microbiology or biostatistics. Unable to carry out critical thinking.

40%-49%: Demonstrate a limited knowledge of microbiology applications and biostatistics. May have some difficulty with critical thinking.

50%-59%: A good understanding of analytical microbiology and topical and emerging microbiological issues with demonstration of some critical thinking.

60%-69%: In addition to the above, good critical thinking skills and showing an ability to interpret statistical analysis of microbiology data.

70%-100%: A highly developed and extensive understanding of analytical microbiology and topical and emerging microbiological issues.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	36	
Practical	24	
Independent Learning	75	

Essential Material

- "Australian government." Guidelines for sterility testing. <http://www.tga.gov.au/docs/pdf/sterilit.pdf>
- "Moodle." www.moodle.wit.ie
- Armstrong, R.A. and A.C. Hilton. _Statistical Analysis in Microbiology: StatNotes_. USA: Wiley-Blackwell, 2014.
- Collins, C.H. _Collins and Lyne's microbiological methods_. 8th Edition. London: Arnold, 2003.
- Madigan, M.T. _Brock Biology of Microorganisms_. 14th Edition. USA: Pearsen , 2014.
- Zar, J.H. _Biostatistical Analysis_. 5th ed.. USA: Prentice Hall, 2007.

Supplementary Material

- Bisen, S.P. _Microbes: concepts and applications_. USA: Wiley-Blackwell, 2012.

Requested Resources

- Lecture Room: Loose Seated
- Room Type: Biology Lab